SUNRISE PRIVATE BANK LTD.

Technology & Digital Banking Division

BUSINESS REQUIREMENTS DOCUMENT

Bank Loan Application System (BLAS)

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Version	1.0 — Final Draft
Prepared By	Business Analysis Team
Reviewed By	Project Steering Committee
Approved By	CTO & Head of Retail Banking
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Status	Approved for Development

1. Document Control & Revision History

Version	Date	Author	Description of Change
0.1	10 Mar 2025	Priya Sharma, Sr. BA	Initial draft — scope and objectives
0.5	28 Mar 2025	Priya Sharma, Sr. BA	Added FR, NFR, stakeholder sections
0.8	14 Apr 2025	Anand Mehta, Tech Lead	Technical review — NFR and constraints
0.9	25 Apr 2025	Steering Committee	Incorporated review comments
1.0	05 May 2025	Priya Sharma, Sr. BA	Final approved version

1.1 Distribution List

Name / Role	Department	Access Level
CTO	Technology	Full
Head of Retail Banking	Retail Banking	Full
Project Manager	PMO	Full
Development Team Lead	Technology	Full
QA Manager	Technology	Full
Compliance Officer	Legal & Compliance	Read-Only
Internal Auditor	Audit	Read-Only

2. Executive Summary

Sunrise Private Bank Ltd. (hereinafter referred to as "the Bank") is a mid-sized Schedule Commercial Bank headquartered in Mumbai, Maharashtra, with a network of 148 branches across 12 states in India. The Bank currently processes retail and MSME loan applications primarily through paper-based and semi-digital workflows, resulting in high processing costs, extended turnaround times, inconsistent credit decisions, and poor customer experience.

The Bank's Board of Directors and Executive Committee have approved the development and deployment of a comprehensive Bank Loan Application System (BLAS) — a unified, end-to-end digital platform that will automate and standardise the entire loan lifecycle from online application capture to disbursal and post-disbursement monitoring.

This Business Requirements Document (BRD) defines the complete set of business requirements, functional specifications, non-functional expectations, assumptions, and constraints that must be addressed by the BLAS solution. It serves as the primary contractual reference between business stakeholders and the technology delivery team throughout the project lifecycle.

2.1 Problem Statement

The Bank's existing loan processing environment suffers from the following critical pain points:

- Average loan processing time of 12–18 business days for retail loans (industry benchmark: 3–5 days).
- Manual document collection and physical verification leading to high re-work rates (~35%).
- Lack of real-time application status visibility for customers and relationship managers.
- Siloed systems for credit scoring, KYC verification, and disbursement with no single source of truth.
- Regulatory compliance lapses due to manual interpretation of RBI, CIBIL, and bureau guidelines.
- High operational costs — estimated INR 4,200 per loan application on average.

2.2 Strategic Objectives

- Reduce average retail loan TAT (Turn-Around Time) to under 5 business days.
- Achieve 80% straight-through processing (STP) rate within 18 months of go-live.
- Reduce cost-per-application by 60% through automation.
- Ensure 100% compliance with RBI Master Directions on KYC and Fair Lending practices.
- Improve Customer Satisfaction Score (CSAT) for lending journeys from 3.1 to 4.5+ (out of 5).
- Enable the Bank to scale loan book from INR 3,200 Cr to INR 6,000 Cr within 3 years.

3. Project Overview

3.1 Project Background

The BLAS project is a Tier-1 strategic initiative under the Bank's 3-Year Digital Transformation Roadmap ("Project Aakash"). It is funded under the FY 2025-26 Capital Expenditure budget with an approved outlay of INR 18.5 crore (including implementation, integration, and two-year support). The project will be delivered in three phases over 18 months.

3.2 Project Phases

Phase	Scope	Target Go-Live	Duration
Phase 1 — Core Platform	Retail loan origination, digital KYC, credit bureau integration, basic workflow	Q3 FY26 (Oct 2025)	6 months
Phase 2 — Advanced Credit & MSME	MSME loans, automated credit scoring engine, advanced rule engine	Q1 FY27 (Jan 2026)	4 months
Phase 3 — Analytics & AI	Predictive default models, portfolio analytics, customer 360 dashboard	Q3 FY27 (Oct 2026)	8 months

3.3 Loan Products in Scope

Product Code	Loan Product	Ticket Size Range	Max Tenure
HL-001	Home Loan	INR 10 Lakh – INR 5 Crore	30 Years
PL-001	Personal Loan	INR 50,000 – INR 30 Lakh	7 Years
AL-001	Auto Loan (New)	INR 1 Lakh – INR 50 Lakh	8 Years
AL-002	Auto Loan (Used)	INR 50,000 – INR 20 Lakh	5 Years
GL-001	Gold Loan	INR 10,000 – INR 50 Lakh	3 Years
EL-001	Education Loan	INR 2 Lakh – INR 40 Lakh	15 Years
MSME-001	MSME Term Loan	INR 5 Lakh – INR 5 Crore	10 Years
MSME-002	Working Capital Loan	INR 1 Lakh – INR 3 Crore	12 Months

4. Project Scope

4.1 In-Scope

4.1.1 Application & Origination

- End-to-end online loan application portal (web and mobile responsive) for retail and MSME customers.
- Multi-channel application capture: Branch, Online Portal, Mobile App, and Partner APIs.
- Intelligent digital document upload with Optical Character Recognition (OCR) for auto-data extraction.
- Application pre-screening and eligibility check before formal submission.

4.1.2 KYC & Identity Verification

- Aadhaar-based eKYC via UIDAI API integration for individual applicants.
- PAN verification via NSDL/UTIITSL API.
- Video KYC (VKYC) module compliant with RBI circular on V-CIP (Video Based Customer Identification Process).
- Business KYC via MCA21 API for MSME entities.

4.1.3 Credit Assessment

- Real-time credit bureau pull from CIBIL TransUnion, Experian, Equifax, and CRIF HighMark.
- Configurable credit scorecard and rule engine for automated credit decisioning.
- Debt-to-Income (DTI) and Fixed Obligation to Income Ratio (FOIR) calculations.
- Negative List / Watchlist screening (RBI defaulter list, CERSAI, WILFUL defaulters).

4.1.4 Workflow & Approval Engine

- Multi-level configurable approval workflows (Branch Manager, Zonal Credit, Central Credit).
- Role-based access control (RBAC) for all workflow actors.
- Deviation management with maker-checker controls and full audit trail.
- SLA tracking and escalation management with automated notifications.

4.1.5 Legal, Valuation & Documentation

- Integration with empanelled legal vetting agencies and property valuers for home loans.
- Auto-generation of sanction letters, loan agreements, and hypothecation deeds using templating engine.
- Digital signature integration (DSC/eSign via MeitY-licensed providers).

4.1.6 Disbursement

- Integration with Core Banking System (CBS — Finacle 10.x) for loan account creation and disbursement.
- NEFT/RTGS/IMPS disbursement initiation via RBI-approved payment infrastructure.
- Tranche disbursement management for construction-linked home loans.

4.1.7 Post-Disbursement & Monitoring

- EMI schedule generation and communication to borrowers.

- Early Warning Signal (EWS) dashboard for portfolio managers.
- Loan modification, top-up, and foreclosure request management.

4.1.8 Reporting & Compliance

- Automated regulatory reports: CRILC (RBI), SMA classification, NPA provisioning.
- MIS dashboards for senior management and business teams.
- Full audit log for all system actions compliant with RBI IT Framework guidelines.

4.2 Out-of-Scope

The following items are explicitly excluded from the current project:

- Corporate / Large Syndicated Loan origination (separate project — Project Udaan).
- Treasury and Investment products (Mutual Funds, Fixed Deposits, Insurance).
- Collections, NPA recovery, and SARFAESI proceedings management.
- Credit card origination and management systems.
- Development of the Core Banking System (CBS); integration only is in scope.
- Customer-facing mobile app development (Mobile Banking team will integrate BLAS APIs).
- International / FOREX loan products.
- Infrastructure provisioning and data centre setup (to be managed by IT Operations).

5. Stakeholder Register

5.1 Stakeholder List & Roles

ID	Stakeholder	Role / Title	Interest & Influence	Engagement Level
S01	Mr. Rajiv Malhotra	Chief Technology Officer (CTO)	Project Sponsor; final approval authority on technical decisions	Champion
S02	Ms. Deepa Krishnan	Head of Retail Banking	Business Sponsor; defines product requirements and accepts deliverables	Champion
S03	Mr. Suresh Patel	Chief Risk Officer (CRO)	Defines credit risk appetite; approves rule engine parameters	Supportive
S04	Ms. Anjali Singh	Chief Compliance Officer	Ensures RBI/regulatory alignment; signs off on compliance controls	Supportive
S05	Mr. Arun Nair	Head of MSME Banking	Requirements owner for MSME loan products	Supportive
S06	Ms. Priya Sharma	Senior Business Analyst (BA Lead)	Requirements elicitation, documentation, and sign-off facilitation	Champion
S07	Mr. Anand Mehta	Technology Lead / Solution Architect	Technical design, integration architecture, non-functional requirements	Champion
S08	Mr. Vikram Joshi	Project Manager	Delivery planning, risk management, stakeholder communication	Champion
S09	Ms. Ritu Bose	Head of Credit — Retail	Defines credit policy rules, underwriting criteria, and approval hierarchies	Supportive
S10	Branch Operations Team	Branch Managers & Relationship Managers	End users of the system; process loan applications on behalf of walk-in customers	Neutral
S11	Mr. Kiran Desai	Head of Internal Audit	Reviews audit controls; independent assurance on compliance	Neutral
S12	External Customers	Retail & MSME Loan Applicants	End beneficiaries; use the customer-facing portal and mobile interface	Neutral
S13	RBI / UIDAI / CIBIL	Regulatory & Statutory Bodies	Mandate compliance requirements; provide data APIs (bureau, KYC)	Neutral
S14	Infosys (CBS Vendor)	Core Banking Vendor	Provides API support for Finacle 10.x CBS integration	Supportive
S15	QA & Testing Team	QA Manager + Testers	Define test strategy, execute UAT, sign off on quality	Supportive

6. Functional Requirements

Each functional requirement is documented in a structured format. Priority levels: High (must have for Phase 1 go-live), Medium (required by Phase 2), Low (Phase 3 enhancement).

FR-001 | Multi-Channel Loan Application Capture	
Description	The system shall enable loan applicants to initiate and submit loan applications through multiple channels: (a) Customer self-service web portal, (b) Mobile-responsive web interface, (c) Branch Officer-assisted entry, and (d) Partner/DSA (Direct Sales Agent) API channel. All channels must converge into a single unified application repository.
Priority	High
Primary Actors	Retail Customers, MSME Customers, Relationship Managers, DSA Partners
Inputs	Applicant personal details, loan amount, product selection, channel identifier, referral code
Outputs	Unique Application Reference Number (ARN), acknowledgement SMS and email, application dashboard entry
Business Rules	ARN must be system-generated and unique across all channels. Partial applications must be saveable and retrievable for up to 30 days. Applications submitted after 6:00 PM IST shall be treated as next business day submissions for SLA purposes.

FR-002 | Digital KYC and Identity Verification	
Description	The system shall perform fully digital customer identification and verification in compliance with the RBI Master Direction on KYC (2016, updated 2023). It shall support Aadhaar OTP-based eKYC, Aadhaar Biometric eKYC, PAN verification, and Video KYC (VKYC) as defined in RBI circular RBI/2021-22/67. Business entities must be verified via MCA21 API.
Priority	High
Primary Actors	Customers, KYC Officers, Compliance Team
Inputs	Aadhaar number, PAN card, consent OTP, liveness video feed, GST certificate, CIN (for companies)
Outputs	KYC status (Verified / Pending / Rejected), KYC score, FATF risk category, masked Aadhaar data stored per UIDAI guidelines
Business Rules	eKYC consent must be captured digitally and stored immutably. PAN must be seeded with Aadhaar as per IT Act requirements. VKYC must be conducted by certified Bank officer. KYC records must be retained for 10 years post account closure per PMLA 2002.

FR-003 | Automated Document Management with OCR	
Description	The system shall enable applicants and Bank officers to upload supporting documents in supported formats (PDF, JPG, PNG). An OCR engine shall automatically extract key data fields from uploaded documents (e.g., name, address, income, account numbers from bank statements) and pre-populate application fields, flagging discrepancies for human review.
Priority	High
Primary Actors	Customers, Relationship Managers, Credit Officers

Inputs	Scanned income proof, bank statements (6 months), ITR, property documents, business financials
Outputs	Extracted data fields, document quality score, checklist completion status, exception flags
Business Rules	Document file size must not exceed 5 MB per file. OCR accuracy must be minimum 92% for standard document types. Rejected/incomplete documents must trigger automated re-upload requests with specific instructions. Original documents must be retained in encrypted form.

FR-004 | Credit Bureau Integration & Score Retrieval	
Description	The system shall perform automated, real-time credit bureau inquiries upon application submission. It must integrate with all four RBI-licensed credit information companies: CIBIL TransUnion, Experian India, Equifax India, and CRIF HighMark. The system shall retrieve credit scores, DPD (Days Past Due) history, outstanding obligations, and enquiry footprint for all individual applicants and guarantors.
Priority	High
Primary Actors	Credit Underwriters, Credit Risk Team
Inputs	Applicant PAN, date of birth, name, address
Outputs	Credit score (300–900), credit report summary, DPD history (24 months), total EMI obligations, number of recent inquiries
Business Rules	Bureau pulls must be logged with timestamp and stored against the application. In case of bureau unavailability, system must queue the request and retry. Consent for bureau inquiry must be captured. Bureau hit charges to be tracked per application for cost reporting.

FR-005 | Automated Eligibility & Pre-screening Engine	
Description	Before formal credit assessment, the system shall perform an automated pre-screening of all applications against configurable eligibility criteria including minimum age (21 years), maximum age at loan maturity (70 years), minimum income thresholds, employment type (salaried/self-employed/business), residential status, CIBIL score floor, and negative list check. Applications failing pre-screening shall be auto-declined with reason codes.
Priority	High
Primary Actors	System (Automated), Customers
Inputs	Application data, credit bureau report, income documents
Outputs	Eligibility status (Pass/Refer/Decline), reason codes, preliminary loan amount and EMI indicative offer
Business Rules	Eligibility rules must be configurable by the Credit Risk team without system code changes. Decline reason codes must comply with RBI Fair Practices Code. Auto-declined applications must be communicated within 24 hours. Minimum 95% accuracy on eligibility decisions measured monthly.

FR-006 | Configurable Multi-Level Approval Workflow	
Description	The system shall support a fully configurable, multi-level loan approval workflow. Approval tiers (Branch, Zonal, Central Credit Committee) shall be triggered based on loan amount, product type, risk grade, and policy deviations. Each approval level shall

	have defined authority matrices, and the system must enforce these with digital maker-checker controls. All approval and rejection decisions must be captured with justifications.
Priority	High
Primary Actors	Branch Managers, Credit Managers, Zonal Credit Officers, Credit Committee
Inputs	Application file, credit assessment report, collateral valuation, policy deviation flags
Outputs	Approval/rejection decision, conditions precedent, sanction terms, audit log entry
Business Rules	No single officer can both initiate and approve a loan. Approval authority limits must be configurable per role. Loans above INR 1 Crore require minimum 2-person credit committee approval. System must enforce 4-eyes principle (maker-checker) on all sanction documents.

FR-007 | Sanction Letter & Loan Agreement Generation	
Description	Upon credit approval, the system shall automatically generate customised sanction letters and loan agreement documents using a pre-approved template engine. Documents shall include all mandated disclosures as per RBI Fair Practices Code and shall be digitally signed by both the Bank officer and the customer using Aadhaar eSign or DSC (Digital Signature Certificate).
Priority	High
Primary Actors	Customers, Credit Officers, Legal Team
Inputs	Approved sanction terms, customer KYC data, product-specific legal template, eSign consent
Outputs	Digitally signed sanction letter (PDF), loan agreement, MITC (Most Important Terms and Conditions) document
Business Rules	Sanction letters must be generated within 2 hours of credit approval. All documents must be archived in the Document Management System (DMS) with version control. Tamper-evident digital signatures using MeitY-licensed TSP (Trust Service Provider). Customer copy must be emailed within the same business day.

FR-008 | CBS Integration for Account Creation & Disbursement	
Description	The system shall integrate bi-directionally with the Bank's Core Banking System (Infosys Finacle 10.x) to (a) create loan accounts with correct product parameters upon sanction, (b) generate repayment schedules, (c) initiate disbursement transactions via NEFT/RTGS/IMPS, and (d) receive real-time CBS confirmation of fund credit. Tranche-based disbursement must be supported for home loans with construction linkage.
Priority	High
Primary Actors	Disbursement Officers, CBS Admin, Customers
Inputs	Approved sanction terms, beneficiary bank account details, disbursement schedule
Outputs	Loan account number, disbursement confirmation with UTR number, repayment schedule, CBS transaction ID
Business Rules	CBS integration must use secured REST/SOAP APIs with mutual TLS authentication. Disbursements above INR 2 Lakh must use RTGS. Failed disbursements must trigger automated alerts and be retried maximum 3 times before escalation. Reconciliation reports to be auto-generated daily.

FR-009 | Real-Time Application Status Tracking	
Description	The system shall provide real-time, stage-wise application status tracking accessible to customers (via web/SMS/email), Relationship Managers, and Supervisors. Status updates must be triggered automatically at each workflow stage transition. Customers must be able to track their application using only the ARN and registered mobile number without requiring login credentials.
Priority	High
Primary Actors	Customers, Relationship Managers, Branch Managers
Inputs	Application Reference Number (ARN), registered mobile number
Outputs	Current application stage, pending actions, estimated completion date, contact details for queries
Business Rules	Status SMS must be sent within 5 minutes of stage transition. Email notifications must include next steps and required actions. Status portal must be available 99.9% of the time. Status information must not reveal confidential credit parameters to the applicant.

FR-010 | SLA Management & Escalation Engine	
Description	The system shall define, enforce, and track SLAs for every stage of the loan application workflow. Configurable SLA timers must be activated upon entry to each stage. Near-breach warnings (at 75% of SLA time) and breach alerts must be sent to the responsible officer and their supervisor. Breached applications must auto-escalate to the next level of management with a dashboard flag.
Priority	Medium
Primary Actors	All workflow actors, Branch Managers, Operations Head
Inputs	Stage entry timestamp, SLA configuration, escalation matrix
Outputs	SLA status (On-Track / At-Risk / Breached), escalation notifications, SLA compliance report
Business Rules	SLA timers must pause during awaited customer actions (pending documents, pending eSign). SLA configurations must be editable by Operations without developer intervention. SLA compliance must be reported weekly to the Operations Committee. Regulatory mandated timelines (e.g., 30-day acknowledgement under CPA) must override internal SLA configurations.

FR-011 | Regulatory Reporting & Compliance Module	
Description	The system shall automatically generate all mandatory regulatory reports and returns including: (a) CRILC (Central Repository of Information on Large Credits) monthly submission to RBI for exposures above INR 5 Crore, (b) SMA (Special Mention Account) classification and reporting, (c) CERSAI registration for mortgage loans, (d) CKYC (Central KYC Registry) upload, and (e) GST TDS certificates for applicable products.
Priority	High
Primary Actors	Compliance Team, Finance Team, RBI
Inputs	Loan portfolio data, customer master data, payment history
Outputs	Machine-readable regulatory returns in prescribed formats, submission acknowledgements, audit trail of submissions

Business Rules	CRILC reports must be generated by 10th of each month. System must validate data before submission and flag errors. All submissions must be logged with timestamps, authorisation trail, and acknowledgement receipts. System must be updated within 30 days of any change in regulatory formats.

FR-012 | Early Warning Signal (EWS) Dashboard	
Description	The system shall implement an Early Warning Signal engine that monitors borrower accounts post-disbursement for pre-defined stress indicators including: irregular repayment patterns, bounce in EMI mandates (NACH), significant drop in account balance, deterioration in credit bureau score, legal proceedings filings, and industry/sector stress signals. Flagged accounts shall appear on a dedicated EWS dashboard for portfolio managers.
Priority	Medium
Primary Actors	Portfolio Managers, Credit Risk Team, CRO
Inputs	CBS repayment data, NACH returns, bureau refresh data, external news feeds
Outputs	EWS dashboard with risk-categorised account list, signal details, recommended actions, escalation trail
Business Rules	EWS scoring model parameters must be approved by CRO. Level-1 signals must be reviewed within 3 business days. Accounts with 3 or more active signals must be automatically referred for relationship review. EWS data must be retained for 7 years for regulatory inspection.

FR-013 | Loan Modification, Top-Up & Foreclosure Management	
Description	The system shall support post-disbursement loan lifecycle events including: (a) interest rate repricing (RLLR/MCLR changes), (b) EMI restructuring or moratorium requests, (c) top-up loan processing against existing eligible loans, (d) partial prepayment processing, and (e) full foreclosure with automated NOC (No Objection Certificate) and lien release initiation. All modifications must follow a configurable approval workflow.
Priority	Medium
Primary Actors	Existing Borrowers, Relationship Managers, Credit Officers
Inputs	Modification request type, supporting financial documents, CBS account details
Outputs	Revised loan terms, modified repayment schedule, CBS amendment confirmation, NOC document (for foreclosure)
Business Rules	Top-up eligibility must be auto-calculated based on repayment track record and current valuation. Foreclosure charges must be computed per RBI Fair Practices Code (no foreclosure penalty on floating rate retail loans). Lien/charge release must be initiated within 7 working days of full repayment per RBI guidelines.

FR-014 | MIS & Analytics Dashboards	
Description	The system shall provide role-specific, interactive Management Information System (MIS) dashboards with real-time and periodic analytics. Dashboards shall include: (a) Branch-level application pipeline and conversion funnel, (b) Product-wise sanction and disbursement volumes, (c) Turnaround time analysis by product and channel, (d) Credit bureau score distribution of sanctioned portfolio, (e) Rejection reason analysis, and (f) EWS portfolio health indicators.
Priority	High

Primary Actors	Branch Managers, Business Heads, CTO, MD & CEO
Inputs	All transactional data across the loan lifecycle
Outputs	Interactive dashboards with drill-down, scheduled email reports, exportable Excel/PDF reports
Business Rules	Dashboards must refresh at minimum every 4 hours during business hours and daily at 6:00 AM otherwise. Data used in dashboards must have a documented lineage/source mapping. Dashboard access must be role-restricted. All report downloads must be logged in the audit trail.

7. Non-Functional Requirements (NFRs)

7.1 Performance Requirements

NFR-ID	Requirement	Metric / Threshold	Priority
NFR-P01	Application page load time	< 2 seconds at P95 under normal load	High
NFR-P02	Credit bureau API response	< 5 seconds per bureau hit under normal conditions	High
NFR-P03	Concurrent users	System must support 2,000 concurrent users without degradation	High
NFR-P04	Peak transaction throughput	Minimum 500 loan application submissions per hour	High
NFR-P05	Batch processing (nightly)	Nightly EWS batch and regulatory report generation must complete within 3-hour batch window (1 AM – 4 AM IST)	Medium
NFR-P06	Search response time	Application search results returned in < 1 second for indexed fields	Medium

7.2 Availability & Reliability

NFR-ID	Requirement	Metric / Threshold	Priority
NFR-A01	System availability (SLA)	99.9% uptime (excluding planned maintenance) = max 8.7 hours downtime/year	High
NFR-A02	Planned maintenance window	Maximum 4 hours per month; Sunday 2:00 AM – 6:00 AM IST	High
NFR-A03	Recovery Time Objective (RTO)	Maximum 2 hours for critical system components (CBS integration, eKYC)	High
NFR-A04	Recovery Point Objective (RPO)	Maximum 15 minutes data loss on primary database in event of failure	High
NFR-A05	Disaster Recovery (DR)	Hot standby DR site with automatic failover; DR drill mandatory every 6 months	High
NFR-A06	Data backup	Full daily backup + incremental every 4 hours; retained for 10 years	High

7.3 Security Requirements

NFR-ID	Requirement	Standard / Reference	Priority
NFR-S01	Data encryption at rest	AES-256 encryption for all PII and financial data; FIPS 140-2 compliant modules	High
NFR-S02	Data encryption in transit	TLS 1.2 minimum (TLS 1.3 preferred) for all API and user traffic	High
NFR-S03	Multi-Factor Authentication (MFA)	Mandatory for all Bank officers; OTP on mobile for customer login	High
NFR-S04	Role-Based Access Control	Granular RBAC with principle of least privilege; access reviews every 6 months	High
NFR-S05	Audit logging	Immutable audit log for all data access, modifications, and login events; 7-year retention per RBI IT Framework	High
NFR-S06	VAPT (Vulnerability Assessment)	Mandatory VAPT by certified agency before go-live; annual thereafter; OWASP Top 10 compliance	High
NFR-S07	PII data masking	Aadhaar number to display only last 4 digits; account numbers masked per RBI guidelines	High
NFR-S08	Session management	Auto session timeout after 15 minutes of inactivity for Bank officer interfaces	Medium

7.4 Scalability & Maintainability

- The system must be built on a microservices / API-first architecture to allow independent scaling of components (e.g., OCR service, rule engine, bureau integration).
- The system must support horizontal scaling to handle 3x current projected load within 48 hours with no code changes (infrastructure scaling only).
- Configuration changes to eligibility rules, approval matrices, interest rates, and product parameters must be achievable by authorised business users through a configuration portal without developer involvement.
- The system must provide a full REST API catalogue (OpenAPI 3.0 specification) to enable future integration with partner fintechs and CBS upgrades.
- Code must follow SOLID principles; minimum 80% unit test coverage. All production deployments must pass automated regression suite.

7.5 Usability Requirements

- The customer-facing portal must achieve a System Usability Scale (SUS) score of 75 or above in user testing.

- Application must be fully functional on Chrome 100+, Firefox 100+, Safari 15+, and Edge 100+ browsers.
- Customer portal must be WCAG 2.1 Level AA compliant for accessibility.
- All customer-facing content must be available in English and Hindi. Marathi, Tamil, and Telugu language support to be added in Phase 3.
- Mobile-responsive design must maintain full functionality on screens 375px and above.

7.6 Regulatory & Compliance NFRs

- The system must comply with: RBI Master Direction on KYC (2023); RBI IT Framework for NBFCs/Banks (2011); Information Technology Act 2000 and IT (Amendment) Act 2008; Digital Personal Data Protection Act (DPDPA) 2023; PMLA 2002 and PMLA Rules 2005; Companies Act 2013 (for MSME/corporate borrowers); Consumer Protection Act 2019.
- The system must support generation and submission of regulatory returns in formats prescribed by RBI, UIDAI, CERSAI, MCA, and CKYC without manual intervention.
- All customer consent records must be stored with timestamp, IP address, and version of consent text, compliant with DPDPA 2023.

8. Assumptions & Constraints

8.1 Assumptions

The following assumptions have been made in preparing this document. If any assumption proves incorrect, the affected requirements and timelines must be reviewed by the Project Steering Committee.

ID	Assumption	Owner	Impact if Incorrect
A01	The Bank's CBS (Finacle 10.x) APIs for loan account creation and disbursement are available and Infosys will provide integration support within 4 weeks of project kickoff.	S14 (Infosys)	Phase 1 disbursement functionality delayed; critical path impact
A02	UIDAI Aadhaar eKYC APIs will be accessible via the Bank's registered AUA/KUA licence. UIDAI connectivity and SLA (99.5% uptime) will be maintained throughout the project.	S04 (Compliance)	VKYC must be relied upon exclusively; TAT increases
A03	All four credit bureaus (CIBIL, Experian, Equifax, CRIF) have been contracted by the Bank and API credentials are available before Phase 1 system integration testing.	S03 (CRO)	Bureau-linked credit decisioning unavailable at go-live
A04	The Bank's existing Active Directory (AD) / LDAP infrastructure can be used for RBAC integration and will be made available to the BLAS application team.	S07 (Tech Lead)	Separate identity management may be required; added cost
A05	Dedicated UAT environment with production-equivalent infrastructure will be provisioned by IT Operations at least 6 weeks before each phase go-live.	S08 (PM)	UAT timelines compress; quality risk increases
A06	Business stakeholders (Credit Risk, Compliance, Operations) will be available for requirements workshops at least 8 hours per week during the requirements phase.	S06 (BA Lead)	Requirements may be incomplete; rework risk
A07	The DPDPA 2023 implementing rules will be finalised and notified before Phase 1 go-live; if not, the Bank's current consent management framework will be used as interim.	S04 (Compliance)	Consent module redesign may be required mid-project
A08	Net interest rate and product configuration data will be provided by the Product team in a structured format (Excel/API) at least 8 weeks before Phase 1 go-live.	S02 (Head Retail)	Rate engine cannot be configured; product launch delayed

8.2 Constraints

ID	Constraint Type	Description
C01	Budget	Total project budget is capped at INR 18.5 Crore (capex + opex for 2 years). Any requirement change resulting in >5% cost increase requires Steering Committee approval.

ID	Constraint Type	Description
C02	Timeline	Phase 1 go-live is fixed at October 2025 due to regulatory commitment made by the Bank's MD to RBI supervisory team. This date is non-negotiable without Board approval.
C03	Technology	The solution must be deployable on the Bank's existing on-premises data centre (primary) and DR site. Public cloud deployment is not approved by the Board for core banking-adjacent systems at this time.
C04	Regulatory	All KYC processes must strictly comply with the current RBI Master Direction on KYC. Any deviation requires written approval from the Chief Compliance Officer and RBI liaison.
C05	Data Residency	All customer data and financial transaction data must reside on servers physically located within Indian territory, per RBI Data Localisation requirements (RBI circular RBI/2018-19/153).
C06	Integration	The system must integrate with Finacle 10.x using existing middleware (IBM MQ and MuleSoft ESB). No replacement of existing middleware is permitted within this project.
C07	Vendor	The Bank has an existing enterprise licence for Oracle Database 19c. The BLAS primary database must use Oracle 19c. Polyglot persistence (Redis for caching, Elasticsearch for search) is permitted for non-core data.
C08	Resource	The Bank's internal IT team has a maximum capacity of 8 developers and 3 QA engineers allocated to this project. Any additional resource must be onboarded via approved IT vendor empanelment process.

9. Data Requirements & Integration Architecture

9.1 Key External Integrations

Integration ID	System / Service	Provider	Interface Type	Data Exchanged
INT-001	Aadhaar eKYC	UIDAI (AUA/KUA)	REST API (HTTPS)	Aadhaar OTP auth, demographic data, biometric match result
INT-002	PAN Verification	NSDL / UTIITSL	REST API	PAN number, name, DOB, status
INT-003	CIBIL Credit Bureau	TransUnion CIBIL	REST API	Credit score, full bureau report, payment history
INT-004	Experian Bureau	Experian India	REST API	Credit score, report
INT-005	CRIF HighMark	CRIF India	REST API	Credit score, MFI bureau data
INT-006	Equifax Bureau	Equifax India	REST API	Credit score, report
INT-007	Core Banking System	Infosys Finacle 10.x	REST / IBM MQ	Loan a/c creation, disbursement, repayment schedule
INT-008	NACH Mandate	NPCI	File-based + API	NACH mandate registration and debit authorisation
INT-009	eSign / DSC	eMudhra / Sify (MeitY-licensed)	REST API	Document hash, eSign affixation, certificate chain
INT-010	SMS Gateway	Bank's approved vendor	REST API	OTP, status notifications
INT-011	Email Service	Bank's SMTP relay	SMTP/API	Application confirmations, document delivery
INT-012	CERSAI	CERSAI Portal	Web Service / File	Security interest creation and search for mortgages
INT-013	MCA21 (Business KYC)	Ministry of Corporate Affairs	REST API	Company master data, director details, annual filings
INT-014	CKYC Registry	CERSAI (CKYC)	Batch / API	CKYC upload and search

10. Risk Register

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
R01	CBS API delays from Infosys may hold up Phase 1 disbursement module	Medium	High	Early engagement with Infosys; parallel mock API development for testing
R02	UIDAI API downtime impacts eKYC; applications stuck	Low	High	Fallback to VKYC and offline KYC for branch-assisted cases; SLA monitoring
R03	Credit bureau data inconsistency across 4 bureaus causes decisioning errors	Medium	High	Define bureau hierarchy and dispute resolution logic in credit policy before build
R04	Scope creep from business stakeholders adding new loan products mid-project	High	Medium	Strict change control process; CCB (Change Control Board) approval for all new scope
R05	DPDPA 2023 rules notified mid-project requiring consent module redesign	Medium	Medium	Build consent module modularly; assign legal team to monitor rule gazette notifications
R06	Key BA / architect resource attrition during project	Medium	High	Knowledge transfer documentation updated monthly; two-person knowledge cover for critical roles
R07	Performance issues at go-live due to under-estimated user load	Low	High	Mandatory performance testing with 2.5x expected peak load before go-live
R08	Regulatory audit / RBI inspection finding non-compliance in BLAS before go-live	Low	Critical	Compliance sign-off gate at each phase; Internal Audit involvement in UAT

11. Glossary of Terms

Term / Acronym	Full Form	Definition
ARN	Application Reference Number	A unique system-generated identifier assigned to every loan application at the time of submission, used to track the application through its lifecycle.
AUA / KUA	Authentication User Agency / KYC User Agency	UIDAI-licensed entities authorised to access Aadhaar authentication and eKYC services on behalf of residents with consent.
BLAS	Bank Loan Application System	The end-to-end digital platform being developed under this project to manage the complete loan origination, processing, and post-disbursement lifecycle.
CBS	Core Banking System	The Bank's centralised banking software (Infosys Finacle 10.x) that manages all deposit, loan, and treasury accounts and transactions.
CERSAI	Central Registry of Securitisation Asset Reconstruction and Security Interest	A statutory body maintaining a centralised online registry for security interests created on property and assets in India under the SARFAESI Act 2002.
CIBIL	Credit Information Bureau (India) Limited	India's first and most widely used credit bureau, now branded TransUnion CIBIL, which maintains credit histories and generates credit scores (300–900) for individuals and businesses.
CKYC	Central Know Your Customer	A centralised repository maintained by CERSAI that stores KYC records of customers registered with regulated entities, to avoid duplicate KYC across institutions.
CRO	Chief Risk Officer	The senior executive responsible for overseeing all categories of risk, including credit risk, operational risk, and market risk, across the Bank.
CRILC	Central Repository of Information on Large Credits	An RBI-mandated database to which banks must report credit information on all borrowers with aggregate exposures of INR 5 crore and above, used for systemic risk monitoring.
DMS	Document Management System	A secure electronic repository for storing, retrieving, and managing all documents associated with loan applications, including KYC documents, financial statements, and signed agreements.
DPD	Days Past Due	The number of days a loan payment is overdue beyond its scheduled due date; a key component of credit bureau reports used in underwriting.
DSA	Direct Sales Agent	An individual or entity authorised by the Bank to source and process loan applications on behalf of the Bank in exchange for a commission, operating under the Bank's guidelines and RBI's outsourcing norms.
DTI	Debt-to-Income Ratio	A financial metric calculated as the ratio of total monthly debt obligations to gross monthly income; used to assess a borrower's capacity to service additional debt.

Term / Acronym	Full Form	Definition
eKYC	Electronic Know Your Customer	A digital process of verifying a customer's identity using Aadhaar-based OTP or biometric authentication through UIDAI's API, eliminating the need for physical document submission.
EMI	Equated Monthly Instalment	A fixed payment amount a borrower makes to the Bank on a specified date each month, comprising both principal repayment and interest, until the loan is fully repaid.
ESB	Enterprise Service Bus	Middleware software (MuleSoft in this context) that enables communication and data exchange between different applications in the Bank's IT ecosystem using a standardised messaging infrastructure.
EWS	Early Warning Signal	Pre-defined indicators or triggers that identify potential stress or deterioration in a borrower's financial health or repayment capacity, enabling proactive risk management by the Bank.
FOIR	Fixed Obligation to Income Ratio	A credit underwriting parameter representing the ratio of total fixed monthly financial obligations (including proposed EMI) to gross monthly income; used alongside DTI for credit decisions.
FATF	Financial Action Task Force	An intergovernmental organisation that sets international standards for anti-money laundering (AML) and counter-terrorism financing (CTF); its risk categories are used in KYC classification.
GST	Goods and Services Tax	A comprehensive indirect tax levied on the supply of goods and services in India, relevant in BLAS for processing MSME loans where GST registration and returns are used as income proof.
ITR	Income Tax Return	A form filed with the Income Tax Department of India reporting annual income and tax liability; used as primary income proof for self-employed and business loan applicants.
KYC	Know Your Customer	A regulatory process mandated by RBI under the Prevention of Money Laundering Act (PMLA) requiring banks to verify the identity, address, and financial background of all customers.
MCA21	Ministry of Corporate Affairs Portal Version 21	The Government of India's online registry portal for company filings, director information, and incorporation documents, used for business KYC of corporate and MSME borrowers.
MCLR	Marginal Cost of Funds Based Lending Rate	An RBI-mandated internal benchmark rate used by banks to price floating rate loans, calculated based on marginal cost of funds, operating costs, and tenor premium.
MITC	Most Important Terms and Conditions	A key information document mandated by RBI's Fair Practices Code to be provided to all loan applicants in plain language, outlining all material terms of the loan before execution.

Term / Acronym	Full Form	Definition
MSME	Micro, Small, and Medium Enterprises	Businesses classified under the MSMED Act 2006 based on investment and turnover criteria; a key loan segment for the Bank governed by RBI priority sector guidelines.
MFA	Multi-Factor Authentication	An authentication mechanism requiring users to provide two or more verification factors (e.g., password + OTP, password + biometric) to gain access to the system, enhancing security.
NACH	National Automated Clearing House	An NPCI-operated bulk payment system used by banks to collect recurring payments (such as EMIs) through pre-authorised mandates registered by borrowers.
NEFT	National Electronic Funds Transfer	An RBI-operated nationwide electronic fund transfer system that settles transactions in batches on a deferred net settlement basis; used for loan disbursements.
NOC	No Objection Certificate	A document issued by the Bank upon full repayment of a loan, confirming the closure of the loan account and release of any lien or charge on the borrower's assets.
OCR	Optical Character Recognition	Technology that converts images of typed, handwritten, or printed text (as in scanned documents) into machine-readable text, used in BLAS to auto-extract data from uploaded documents.
PAN	Permanent Account Number	A 10-digit alphanumeric identifier issued by the Income Tax Department of India to individuals and entities, mandatory for financial transactions above threshold limits under the Income Tax Act.
PMLA	Prevention of Money Laundering Act	Indian legislation (2002) requiring banks and financial institutions to maintain records of financial transactions, verify customer identity, and report suspicious transactions to the Financial Intelligence Unit (FIU-IND).
RBAC	Role-Based Access Control	A security framework where system access permissions are assigned based on a user's role within the organisation rather than to individual users, ensuring the principle of least privilege.
RBI	Reserve Bank of India	The central bank and primary regulatory authority for all scheduled commercial banks in India, setting monetary policy, licensing banks, and issuing binding guidelines on lending, KYC, and IT.
RLLR	Repo Linked Lending Rate	An external benchmark-based lending rate linked to RBI's repo rate, mandatorily adopted by banks for all floating rate retail loans from October 2019 per RBI circular.
RTGS	Real Time Gross Settlement	An RBI-operated real-time interbank fund transfer system that settles transactions individually on a gross basis (without netting); used for high-value loan disbursements (typically above INR 2 lakh).

Term / Acronym	Full Form	Definition
SMA	Special Mention Account	An RBI classification for loan accounts showing signs of incipient stress before they become NPAs; SMA-0 (>30 days due), SMA-1 (31-60 days), SMA-2 (61-90 days).
STP	Straight-Through Processing	The automated end-to-end processing of a loan application without manual intervention, enabled by digital KYC, automated credit decisioning, and CBS integration.
TAT	Turn-Around Time	The total elapsed time from submission of a complete loan application to the issuance of the sanction letter or disbursement of funds; a key operational efficiency metric.
UIDAI	Unique Identification Authority of India	The statutory body responsible for issuing Aadhaar numbers and operating the Aadhaar authentication and eKYC infrastructure under the Aadhaar Act 2016.
VAPT	Vulnerability Assessment and Penetration Testing	A two-phase cybersecurity testing methodology: VAPT identifies and quantifies security vulnerabilities (VA) and simulates attacks to validate them and assess their exploitability (PT).
VKYC / V-CIP	Video KYC / Video Based Customer Identification Process	An RBI-permitted digital KYC process (circular RBI/2021-22/67) where a Bank official conducts live video interaction with the customer to verify identity and collect KYC details without physical presence.
WILFUL Defaulter	Wilful Defaulter	A borrower classified by the Bank and reported to credit bureaus for deliberately defaulting on loan obligations despite having the capacity to repay, or diverting loan funds to unauthorised purposes; subject to RBI mandated reporting.

12. Document Approval & Sign-Off

This Business Requirements Document has been reviewed and approved by the following authorised signatories. By signing below, the approvers confirm that this document accurately and completely represents the business requirements for the Bank Loan Application System (BLAS) project.

Role	Name	Signature	Date	Approval Status
Business Sponsor — Retail Banking	Ms. Deepa Krishnan	______________________	________	Approved
Project Sponsor — CTO	Mr. Rajiv Malhotra	______________________	________	Approved
Chief Risk Officer	Mr. Suresh Patel	______________________	________	Approved
Chief Compliance Officer	Ms. Anjali Singh	______________________	________	Approved
Business Analyst Lead	Ms. Priya Sharma	______________________	________	Approved
Technology Lead	Mr. Anand Mehta	______________________	________	Approved
Project Manager	Mr. Vikram Joshi	______________________	________	Approved

END OF DOCUMENT

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